Statistical Inference Memo

The analysis showed that the respondents had an average age of 34.3 years and a median age of 34 years. Age was fairly evenly distributed, with a skewness of value of 0.13. Monthly income was different: the mean income was #59,780.29, while the median was #49,481. Income was strongly right-skewed, with a skewness value of 4.03, showing that a small number of high earners pushed the average upward. For this reason, the median gives a better picture of the typical monthly income.

The 95% confidence interval for overall mean monthly income was #56,679.24 to #62,881.34. For Lagos State, the interval was #53,730.26 to #72,209.36. The two-sample t-test found no significant difference between male and female mean income. Male respondents earned an average of #61,328.82, compared with #58,353.72 for females. The p-value was 0.350, which is above 0.05, so the difference was not statistically significant.

The Chi-square test also found no significant relationship between education level and employment status, with a p-value of about 0.810.

The regression analysis showed that the education level had only a very weak relationship with monthly income. The R2 value was 0.00195, meaning education explained only about 0.2% of the differences in income. Each higher education level was linked with an estimated #1,523.47 increase in monthly income, but this result was not statistically significant because the p-value was 0.297. This does not prove that education causes higher income, since factors such as experience, occupation, age, and location may also affect earnings.

Finally, these results should only be applied to a wider population if the survey sample fairly represents that population. Bias could occur if certain states, age groups, education levels, employment groups were overrepresented or underrepresented.